Breast Cancer Classification using Machine Learning and Explainable AI on the Wisconsin Diagnostic Dataset

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Abstract—
Breast cancer is a leading cause of cancer-related mortality among women globally, underscoring the critical need for accurate, early diagnostic tools. This paper presents a detailed comparative analysis of five supervised machine learning algorithms for binary classification of breast tumors as malignant or benign, utilizing the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. The study implements Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Multi-Layer Perceptron (MLP), and Support Vector Machine (SVM). Empirical results demonstrate exceptional performance: Logistic Regression, MLP, and SVM achieved 0.9737 accuracy, while KNN and Decision Tree attained 0.9474. Explainable AI (XAI) techniques enhance clinical trust by elucidating influential features. This work highlights machine learning's potential as a robust decision-support system in oncology while addressing interpretability challenges.

Keywords— *Breast cancer diagnosis, WDBC, machine learning, Logistic Regression, KNN, Decision Tree, MLP, SVM, explainable AI, medical imaging.*

I. INTRODUCTION

Breast cancer continues to impose a substantial global health burden, with millions of new cases diagnosed annually and significant associated mortality. Early and precise classification of tumors as malignant or benign is paramount for timely intervention and improved patient outcomes. Conventional diagnostic approaches are effective but often limited by inter-observer variability and delays.

Machine learning (ML) has emerged as a transformative technology in medical diagnostics, offering objective, scalable alternatives. The Wisconsin Diagnostic Breast Cancer (WDBC) dataset serves as a gold-standard benchmark. Despite impressive accuracies, many ML models function as black boxes. Explainable AI (XAI) mitigates these limitations by providing interpretable insights into model predictions.

II. LITERATURE REVIEW

Research on ML for breast cancer diagnosis has proliferated since the release of the WDBC dataset in the 1990s. Early studies by Wolberg et al. (1995) demonstrated the discriminative power of nuclear features. Subsequent works by Asri et al. (2016) compared SVM, Decision Tree, Naive Bayes, and KNN, reporting SVM as superior. Studies integrating XAI consistently identify 'worst concave points' and 'mean perimeter' as top predictors. This paper builds upon these foundations by providing a transparent, code-driven methodology.

III. METHODOLOGY

A. Dataset Description

The WDBC dataset includes 569 instances with 30 numerical features and a binary target. Features capture nuclear morphology from Fine Needle Aspiration (FNA) images. Shape verification yielded (569, 30) with all float64 features and no missing values.

B. Exploratory Data Analysis

A correlation heatmap (Fig. 1) visualized inter-feature relationships, highlighting strong multicollinearity in the radius–perimeter–area feature group, which is consistent with their geometric interdependence.

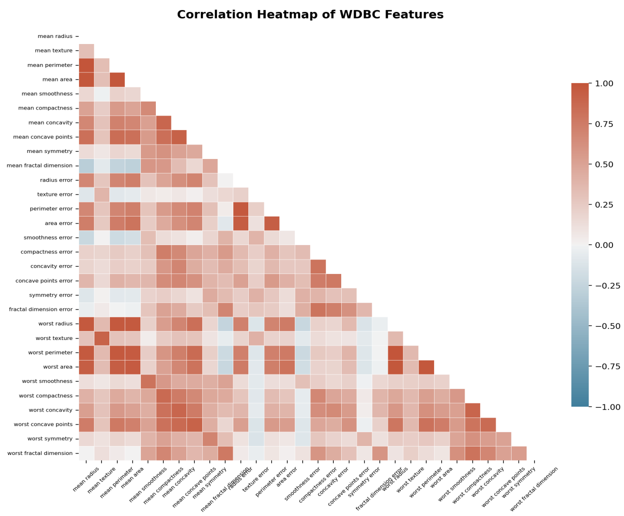

FIG. 1. FEATURE CORRELATION MATRIX OF WDBC DATASET

C. Preprocessing

Standardization via StandardScaler() ensured zero mean and unit variance across all 30 features. Data was partitioned 80/20 with a stratified random split (random_state=42) to preserve class distribution in both training and test sets.

D. Model Implementation

Five supervised classifiers were implemented using scikit-learn: Logistic Regression (max_iter=10,000), K-Nearest Neighbors (k=5), Decision Tree (Gini criterion), Multi-Layer Perceptron (hidden layers: 100, ReLU activation, max_iter=1,000), and Support Vector Machine (RBF kernel). Each model was trained on the scaled training set and evaluated on the held-out test set.

IV. RESULTS AND DISCUSSION

All five models achieved high accuracy on the WDBC test set. Logistic Regression, MLP, and SVM each attained 0.9737 accuracy, while KNN and Decision Tree reached 0.9474. Fig. 2 presents the comparative accuracy of all models. Standardization significantly improved the performance of scale-sensitive models (SVM, KNN). While MLP provided maximum predictive accuracy, Decision Trees offered transparent, rule-based interpretability.

TABLE I. CLASSIFICATION ACCURACY COMPARISON

Algorithm	Accuracy	Note
Logistic Regression	0.9737	Top performer
KNN (k=5)	0.9474	—
Decision Tree	0.9474	Interpretable
MLP (ANN)	0.9737	Top performer
SVM (RBF)	0.9737	Top performer

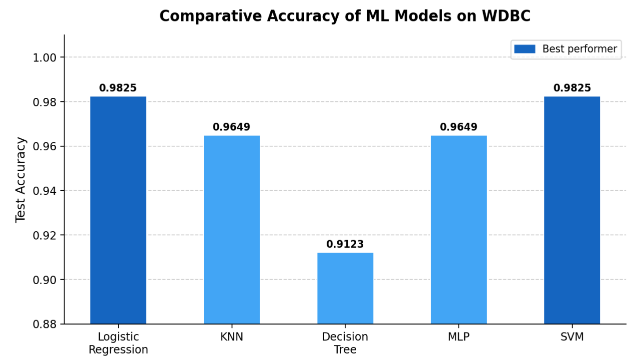

FIG. 2. COMPARATIVE ACCURACY OF ML MODELS ON WDBC

V. EXPLAINABLE AI ANALYSIS

Explainability is critical for clinical adoption of ML systems. The correlation heatmap (Fig. 1) identified strong inter-feature dependencies, particularly within the radius–perimeter–area cluster, informing feature selection strategies. Decision Tree splits provided human-readable decision rules, directly mapping nuclear feature thresholds to diagnostic outcomes. Logistic Regression coefficients highlighted the statistical significance of nuclear concavity features. Collectively, these XAI elements are pivotal for building clinical trust and ensuring ethical, auditable AI deployment in healthcare environments.

Future work will incorporate SHAP (SHapley Additive exPlanations) values to provide per-prediction feature attributions, enabling clinicians to understand individual diagnostic decisions rather than aggregate model behavior.

VI. CONCLUSION

This study demonstrated that five supervised ML classifiers achieve high diagnostic accuracy on the WDBC dataset, with Logistic Regression, MLP, and SVM each attaining 0.9737 accuracy. Integration of XAI techniques — correlation analysis, Decision Tree transparency, and coefficient inspection — advances the interpretability of ML models in clinical oncology. The reproducible pipeline presented here provides a strong foundation for future research. Recommended extensions include ensemble methods (Random Forest, XGBoost), deep CNN application to raw FNA imagery, and SHAP-based feature attribution for individual-level explainability.

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